

NAT & PAT

Course Code: COE 3206

Course Title: Computer Networks



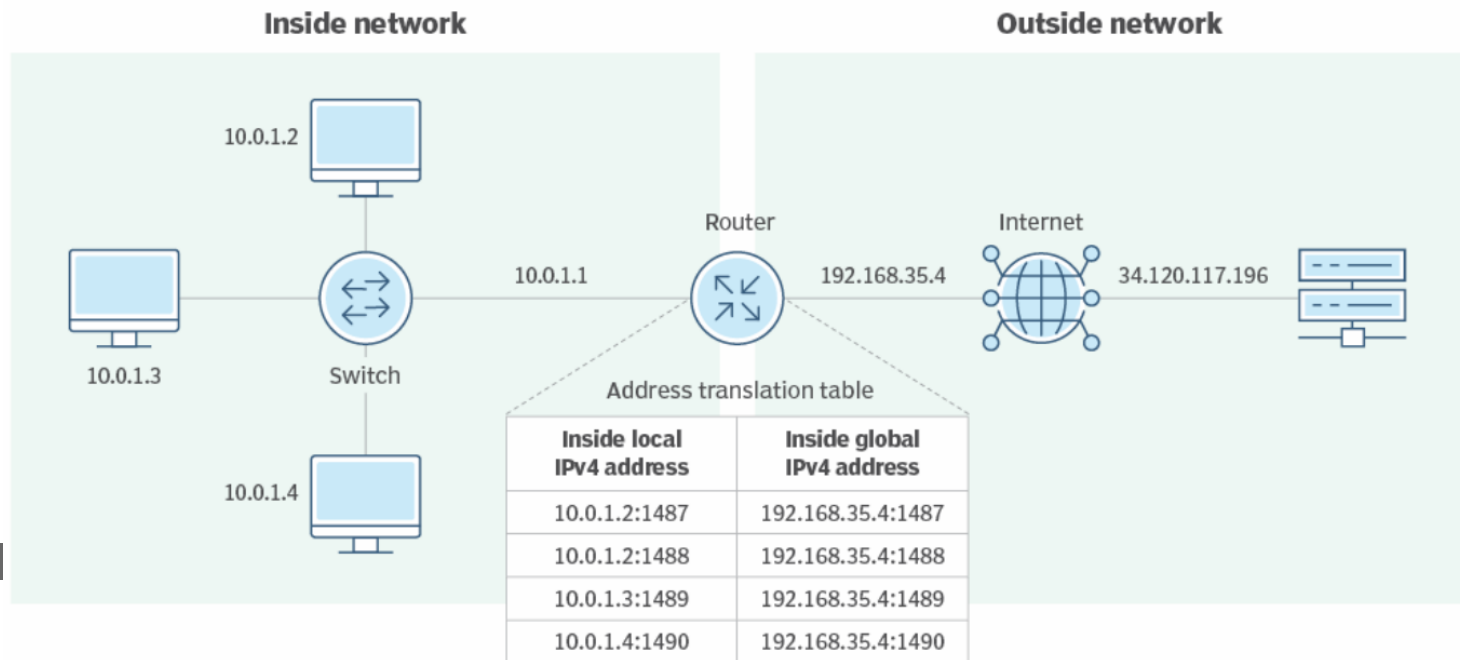
Dept. of Computer Science
Faculty of Science and Technology

Lecturer No:	12	Week No:		Semester:	Fall 25-26
Assistant Professor:	Dr. Rajarshi Roy Chowdhury (rajarshi@aiub.edu)				

Lecture Outline



1. Private Network
2. NAT
3. NAT Operation
4. Pooling of IP Address
5. PAT
6. Load Balancing of Servers
7. Advantage/Disadvantage



Private Network



Private IP network is an IP network that is **not directly connected to the Internet**. IP addresses in a private network can be assigned arbitrarily. **Not registered and not guaranteed to be globally unique.**

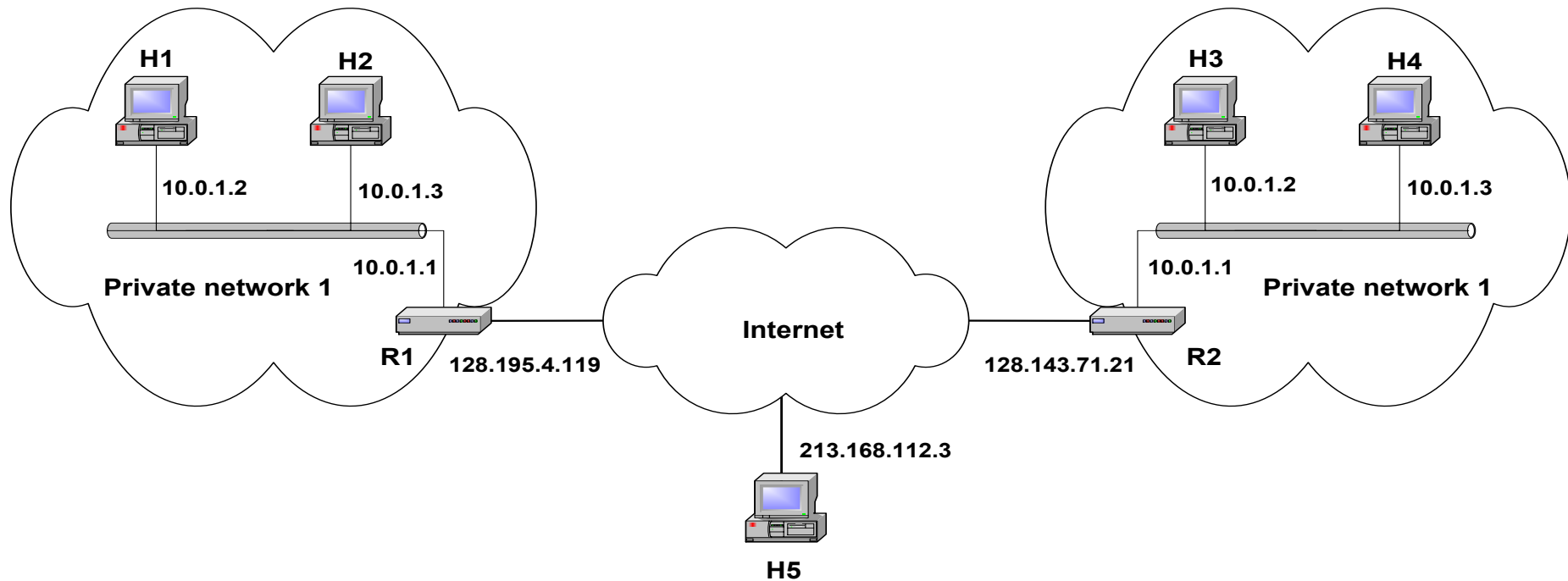
Generally, private networks use addresses from the following experimental address ranges (*non-routable addresses*):

10.0.0.0 – 10.255.255.255/8

172.16.0.0 – 172.31.255.255/16

192.168.0.0 – 192.168.0.255/24

Private Addresses



Network Address Translation (NAT)



A **short-term solution** to the problem of the depletion of IP addresses

- CIDR (Classless Inter Domain Routing) is a possible short-term solution
- NAT is another
- **Long term solution** is IPv6

NAT is a way to conserve IP addresses

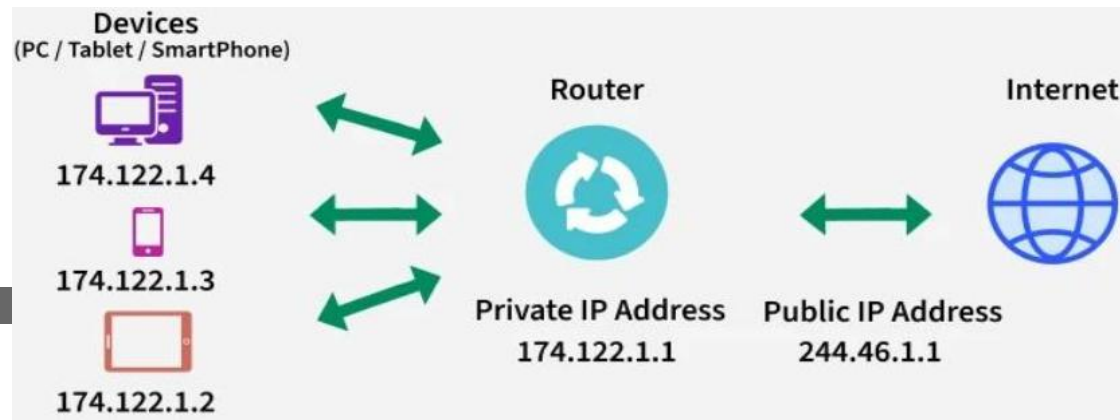
Can be used to hide a number of hosts behind a single IP address

Uses private addresses:

10.0.0.0-10.255.255.255,

172.16.0.0-172.32.255.255 or

192.168.0.0-192.168.0.255



NAT (cont.)

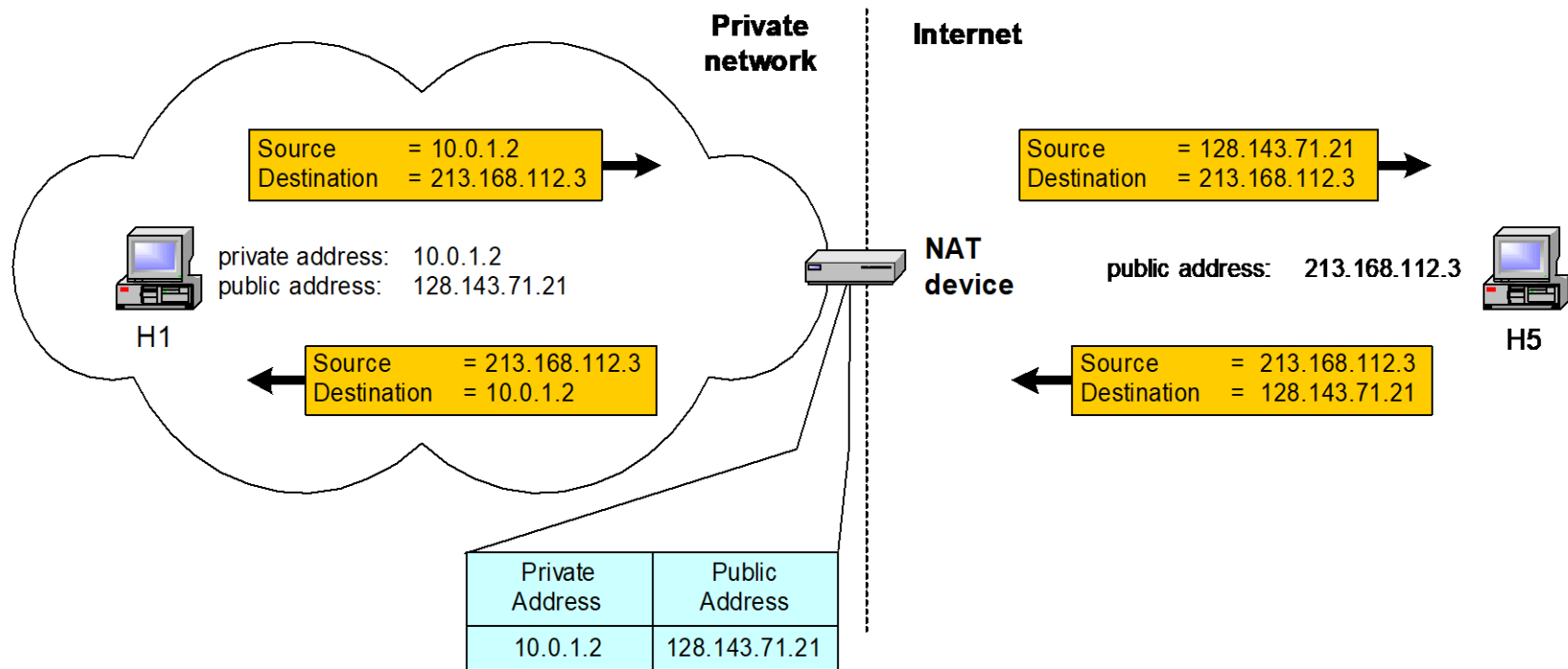


NAT is a **router function** where IP addresses (and possibly port numbers) of IP datagrams are replaced at the boundary of a private network

NAT is a method that enables hosts on private networks to communicate with hosts on the Internet

NAT is run on routers that connect private networks to the public Internet, to replace the IP **address-port pair** of an IP packet with another IP **address-port pair**.

Basic Operation of NAT



NAT device has **address translation table**

One to one address translation

Pooling of IP Addresses

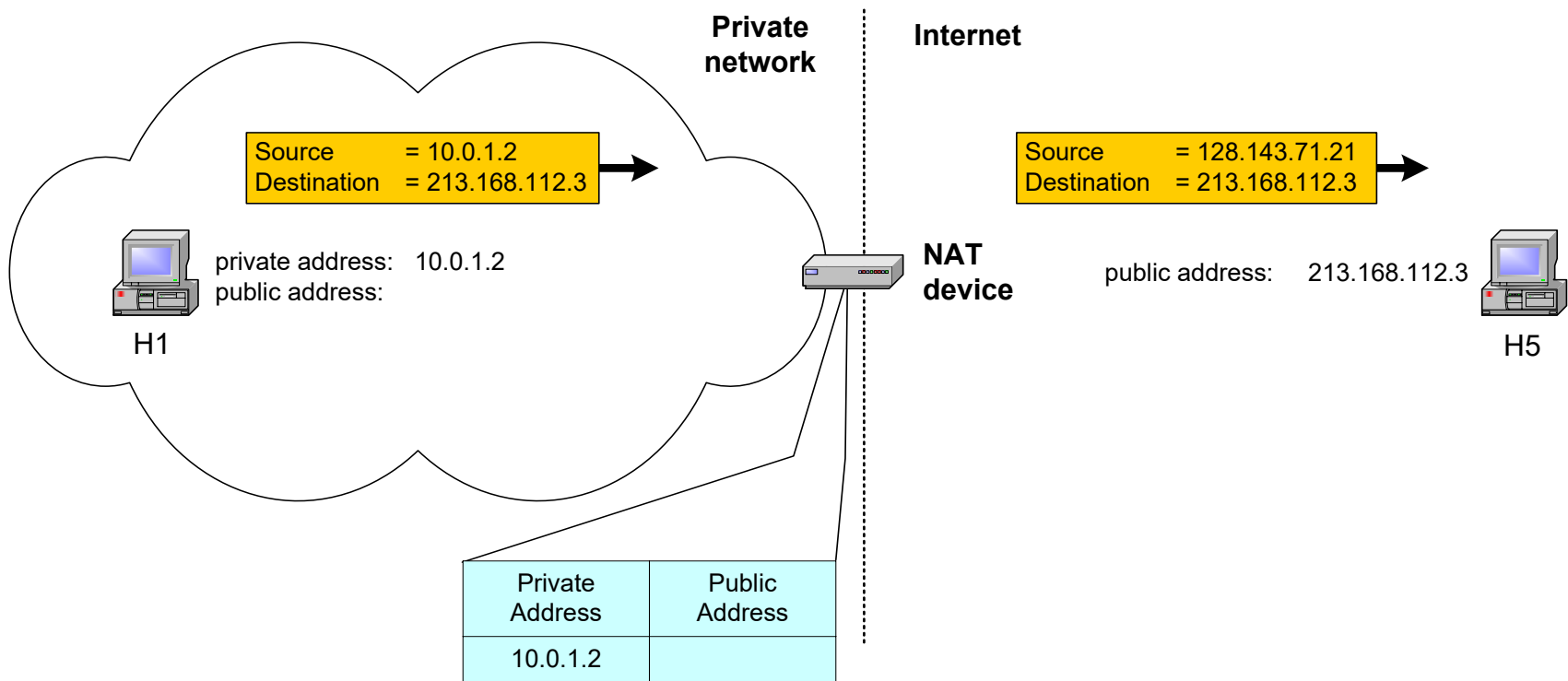


Scenario: Corporate network has many hosts but only a small number of public IP addresses

NAT solution:

- Corporate network is managed with a private address space
- NAT device, located at the boundary between the corporate network and the public Internet, manages a pool of public IP addresses
- When a host from the corporate network sends an IP datagram to a host in the public Internet, the NAT device picks a public IP address from the address pool, and binds this address to the private address of the host

Pooling of IP Addresses (cont...)



Pool of addresses: 128.143.71.0-128.143.71.30

Supporting Migration between Network Service Providers

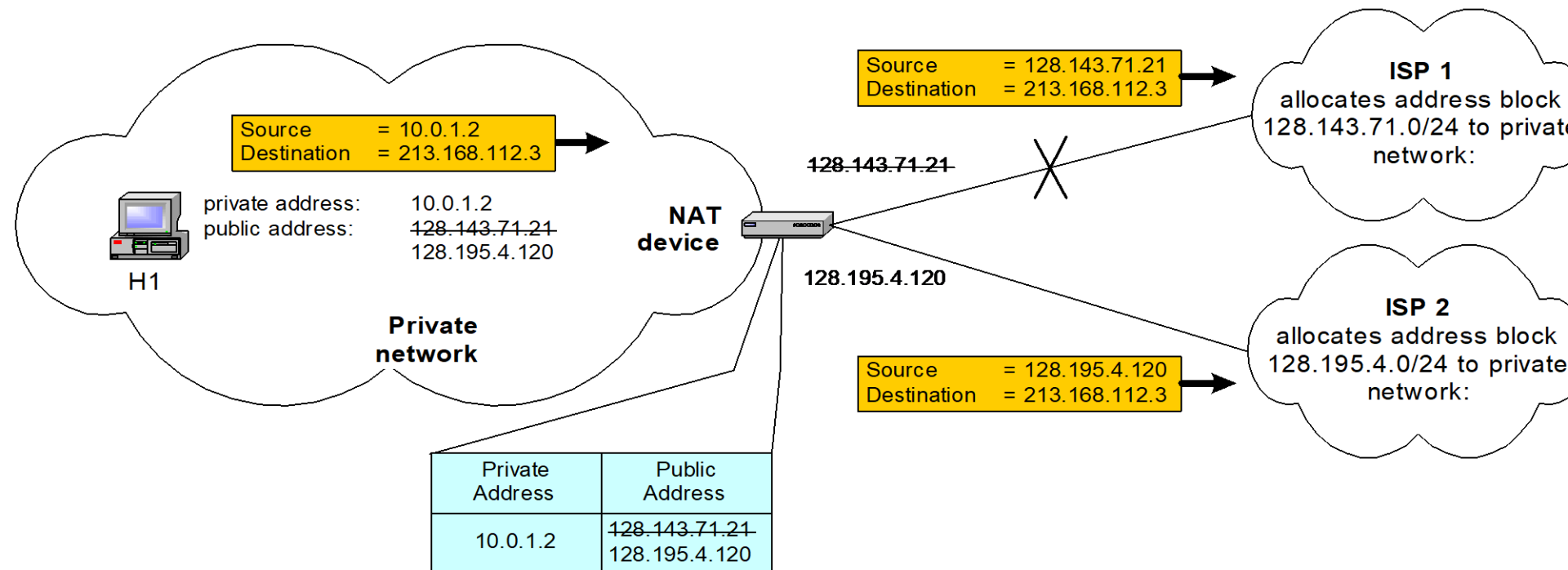


Scenario: In CIDR, the IP addresses in a corporate network are obtained from the service provider. Changing the service provider requires changing all IP addresses in the network.

NAT solution:

- Assign private addresses to the hosts of the corporate network
- NAT device has static address translation entries which bind the private address of a host to the public address.
- Migration to a new network service provider merely requires an update of the NAT device. The migration is not noticeable to the hosts on the network.

Supporting Migration between network service Providers (cont...)



IP Masquerading



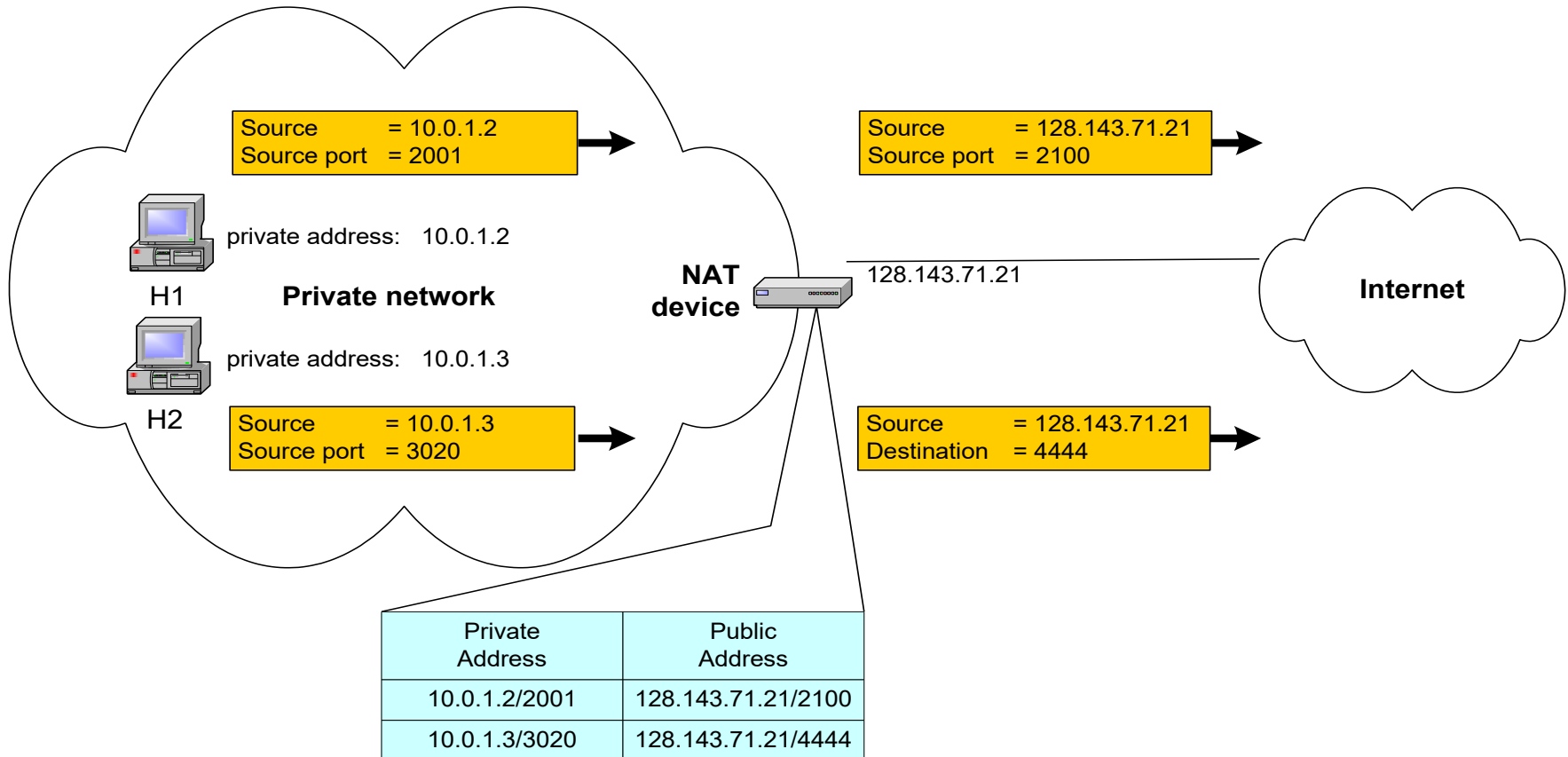
Also called: Network address and port translation (NAPT), port address translation (PAT).

Scenario: Single public IP address is mapped to multiple hosts in a private network.

NAT solution:

- Assign private addresses to the hosts of the corporate network
- NAT device modifies the port numbers for outgoing traffic

IP Masquerading (cont...)



Load Balancing of Servers

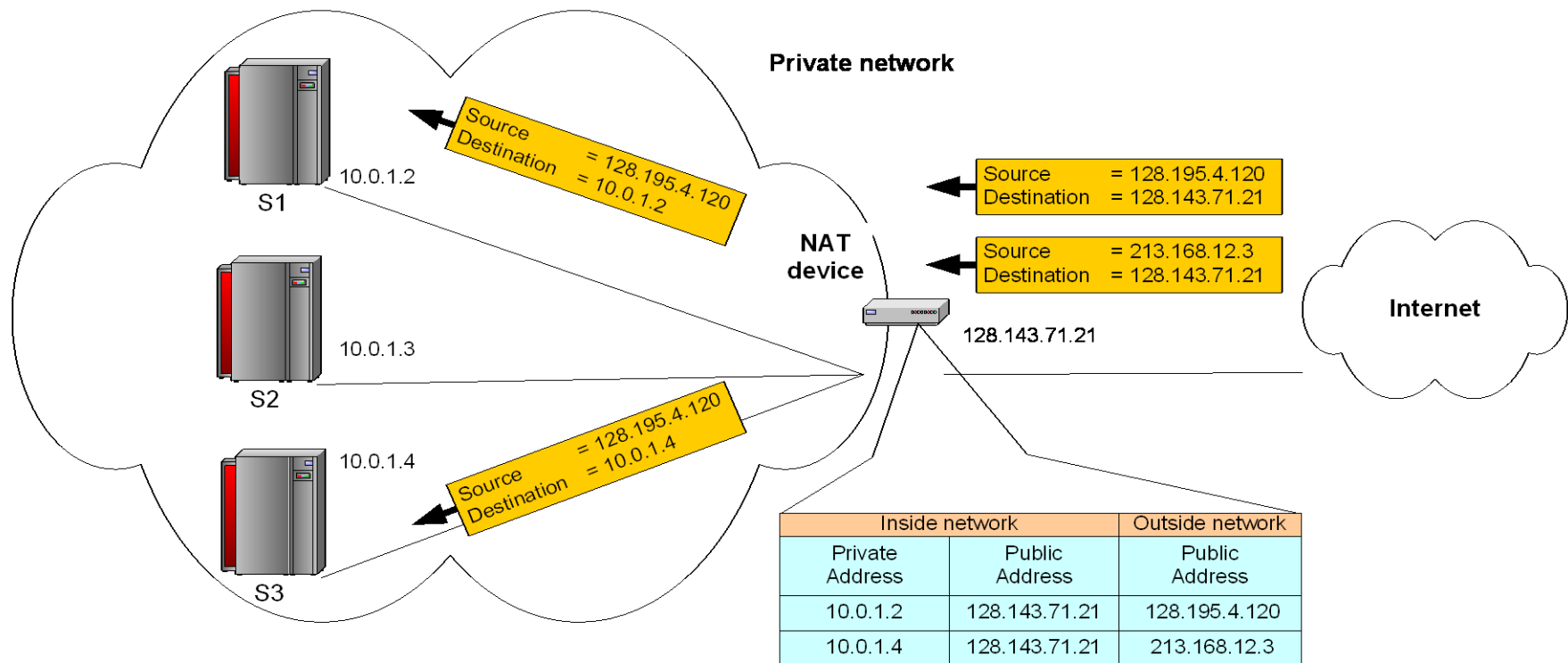


Scenario: Balance the load on a set of identical servers, which are accessible from a single IP address

NAT solution:

- Here, the servers are assigned private addresses
- NAT device acts as a proxy for requests to the server from the public network
- The NAT device changes the destination IP address of arriving packets to one of the private addresses for a server
- A sensible strategy for balancing the load of the servers is to assign the addresses of the servers in a round-robin fashion.

Load Balancing of Servers (cont...)



Concerns about NAT



Performance:

- **Modifying the IP header** by changing the IP address requires that NAT boxes recalculate the IP header checksum
- **Modifying port number** requires that NAT boxes recalculate TCP checksum

Fragmentation

Care must be taken that a datagram that is fragmented before it reaches the NAT device, is not assigned a different IP address or different port numbers for each of the fragments.

Advantage



Advantages of NAT

- The main advantage of NAT is that it can prevent the depletion of IPv4 addresses.
- NAT can provide an additional layer of security by making the original source and destination addresses hidden.
- NAT provides increased flexibility when connecting to the public Internet.
- NAT allows to use your own private IPv4 addressing system and prevent the internal address changes if you change the service provider.

Disadvantage



Disadvantages of NAT

- NAT is a processor and memory resource consuming technology, since NAT need to translate IPv4 addresses for all incoming and outgoing IPv4 datagrams and to keep the translation details in memory.
- NAT may cause delay in IPv4 communication.
- NAT cause loss of end-device to end-device IP traceability



References

1. **Data Communications and Networking**, *B. A. Forouzan*, McGraw-Hill, Inc., Fourth Edition, 2007, USA.
2. <https://www.geeksforgeeks.org/basics-computer-networking/>
3. https://www.tutorialspoint.com/computer_fundamentals/computer_networking.htm



Books

1. **Data Communications and Networking**, *B. A. Forouzan*, McGraw-Hill, Inc., Fourth Edition, 2007, USA.
2. **Computer Networking: A Top-Down Approach**, *J. F., Kurose, K. W. Ross*, Pearson Education, Inc., Sixth Edition, USA.
3. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
4. **CCNA Routing and Switching**, *T. Lammle*, John Wiley & Sons, Second Edition, 2016, USA.
5. **TCP/IP Protocol Suite**, *B. A. Forouzan*, McGraw-Hill, Inc., Fourth Edition, 2009, USA.
6. **Data and Computer Communication**, *W. Stallings*, Pearson Education, Inc., 10th Edition, 2013, USA.